HIKET — Bayesian Calibration

Methods and pipeline documentation

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Contents

1 Overview

This document describes the Bayesian calibration pipeline for the HIKET multi-model SOC intercomparison study. The calibration is implemented in a model-agnostic engine (`calibration_engine.R`) and four model-specific orchestration scripts (`run_{Model}_calibration.R`).

Models in the intercomparison: Yasso07, Yasso15, Yasso20, RothC (pending).

Core research question: Do divergent model projections of Finnish forest carbon sink saturation reflect genuine structural differences between models, or calibration artefacts? All models are calibrated against the same data with the same statistical framework; structural differences are designed to appear in residuals rather than being absorbed into error terms.

Pipeline split:

Script	Purpose
`Data_work.R`	Data preparation
`run_{Model}_calibration.R`	MCMC calibration, posterior samples
`run_{Model}_predictive.R`	Posterior predictive trajectories, obs vs pred
`run_residual_analysis.R`	Random forest residual analysis (model-agnostic)

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2 Models

2.1 Yasso07

The Yasso07 model (Tuomi et al. 2009, 2011) describes SOC decomposition as first-order kinetics in five pools: A (acid-soluble), W (water-soluble), E (ethanol-soluble), N (non-soluble/lignin), and H (humus). Litter enters the A, W, E, N pools proportionally to AWEN fractions; material flows between AWEN pools (controlled by transfer fractions p_{ij}) and to H; H decomposes irreversibly.

The climate modifier is applied multiplicatively to all decomposition rates:

$$\xi(T, T_{\text{amp}}, P) = \exp(\beta_1(T - T_r) + \beta_2(T^2 - T_r^2)) \times (1 - \exp(\gamma P))$$

where T is annual mean temperature, T_{amp} is amplitude (internally decomposed into a sinusoidal within-year cycle), P is annual precipitation, $T_r = 15\,^{\circ}\text{C}$ is the reference temperature, and β_1,

β_2, γ are free parameters. A woody decomposition modifier is applied as an additional factor for FWL/CWL size classes (δ_1, δ_2, r). Model core implemented in Fortran; called via `.Fortran()`.

2.2 Yasso15 / Yasso20

Structurally similar to Yasso07 with pool-group-specific climate responses (separate (β, γ) sets for A/W/E pools, N pool, and H pool). Yasso20 additionally modifies the temperature response shape. Calibration follows the same architecture as Yasso07 with an expanded parameter vector.

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3 Parameter Specification — Yasso07

3.1 Fixed Parameters

Six parameters are fixed at published values (Tuomi et al. 2009, 2011) and are not calibrated:

Parameter	Physical meaning
α_A, α_W, α_E, α_N	Decomposition rate constants for the four litter pools (yr^{-1})
p_H	Fraction of decomposed material entering the humus pool
α_H	Humus decomposition rate (yr^{-1})

Rationale for fixing: Rate constants are well-constrained in the international Yasso literature. Calibrating them risks overfitting to Finnish climate and litter inputs. Structural differences between models are encoded in flow fractions and climate response, the scientifically interesting quantities for this intercomparison.

3.2 Free Parameters

20 parameters are jointly calibrated (18 Yasso07 physical + 2 nuisance):

Inter-pool flow fractions (12): p_{AW}, p_{AE}, p_{AN}, p_{WA}, p_{WE}, p_{WN}, p_{EA}, p_{EW}, p_{EN}, p_{NA}, p_{NW}, p_{NE}. Each donor pool i must satisfy $\sum_j p_{ij} \leq B = 1 - p_H$ (mass balance with humus formation).

Climate and woody response (6): β_1, β_2 (temperature, linear and quadratic), γ (precipitation), δ_1, δ_2, r (woody litter decomposition adjustment).

Nuisance parameters (2):

- σ_{init}: multiplicative initial-condition uncertainty. Added in quadrature to σ_{obs} at the first observation per plot. Captures error from assuming the 1985 SOC equals the analytical steady state.
- σ_{input}: global litter scaling multiplier. Applied to all litter inputs before model evaluation (steady state and transient run consistently). Captures allometric uncertainty in NFI litter estimates (~20–50% intrinsic uncertainty). Currently global (same multiplier for all plots, species, and years).

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4 Parameter Transformations

The DEzs sampler operates in unconstrained $\mathbb{R}^{20}$. Physical parameters are recovered by applying `to_original()` to the sampler's proposals. Three primitive transform types are used.

4.1 Stick-Breaking (flow fractions)

Each of the four donor columns (A, W, E, N) has three outflows satisfying:

$$p_{ij} \geq 0 \quad \forall j, \qquad \sum_j p_{ij} \leq B$$

The stick-breaking transform converts a vector $v \in \mathbb{R}^3$ to a valid fraction vector p:

$$p_i = \sigma(v_i) \times r_i, \qquad r_{i+1} = r_i - p_i, \quad r_1 = B$$

where $\sigma(v) = (1 + e^{-v})^{-1}$ is the logistic sigmoid. The remaining stick r_i shrinks at each step, guaranteeing $\sum_j p_{ij} \leq B$. The inverse (physical $\rightarrow$ unconstrained) follows by sequential logit transformation of the normalised remainder fractions.

4.2 Log Transform (σ_{init}, σ_{input})

$$v = \log(\theta), \qquad \theta = e^v$$

Ensures strict positivity. σ_{input} is centred at 1 in physical space ($v = 0$) by default.

4.3 Unconstrained ($\beta_1, \beta_2, \gamma, \delta_1, \delta_2, r$)

No transformation. These parameters enter the climate modifier through an exponential expression; any real value is physically admissible.

4.4 Jacobian Correction

The log-Jacobian $\log|\partial\theta/\partial v|$ is added to the log-likelihood at every evaluation to correct for the volume distortion:

$$\log p(\theta \mid y) \propto \log p(y \mid \theta) + \log p(\theta) + \log\left|\frac{\partial\theta}{\partial v}\right|$$

For the stick-breaking transform the Jacobian factor per element is $\sigma(v_i)(1 - \sigma(v_i)) \times r_i$. For the log transform it is $e^v = \theta$. The Jacobian is added to the likelihood (not the prior) so that the BayesianTools prior object remains a standard Gaussian.

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5 Prior Distribution

The prior is a multivariate normal in unconstrained space, independent across parameters:

$$p(v) = \prod_{k=1}^{20} \mathcal{N}(v_k;\ v_k^*,\ \sigma_k^2)$$

Prior centre v^*: published Yasso07 defaults (Tuomi et al. 2009, 2011) transformed to unconstrained space. Nuisance parameters centred at $\sigma_{\text{init}} = 0.10$ and $\sigma_{\text{input}} = 1.00$. A round-trip check verifies `to_original(to_unconstrained(theta)) == theta` to machine precision at startup.

Prior widths σ_k (unconstrained space):

Parameter group	σ_k	Rationale
Flow fractions (stick-break)	1.0	Weakly informative; physical range is $[0, B]$
β_1	0.2	$\exp(\beta_1 T)$ at $T \approx 10\,^\circ$C: 95% CI spans $\approx 0.4\times$–$7\times$ modifier
β_2	0.05	Quadratic amplification; tight prior prevents unphysical curvature
$\gamma,\ \delta_1,\ \delta_2,\ r$	1.0	Weakly informative
σ_{init} (log scale)	0.5	Physical 95% CI $\approx$ [0.04, 0.27]: 4%–27% SOC init uncertainty
σ_{input} (log scale)	0.5	Physical 95% CI $\approx$ [0.37, 2.72]: $\approx 2.7\times$ range either way

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6 Likelihood

6.1 Observation Model

A multiplicative-normal (proportional-error) model is used:

$$\text{SOC}_{\text{obs},t} \sim \mathcal{N}(\hat{C}_t,\ s_t^2)$$

where the standard deviation scales with the predicted SOC:

$$s_t = \begin{cases} \hat{C}_t \sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{init}}^2} & \text{first observation per plot (1985)} \\ \hat{C}_t\, \sigma_{\text{obs}} & \text{subsequent observations (2006)} \end{cases}$$

Rationale for multiplicative-normal: A log-normal error model was used during a period when the data contained a $\times 10$ litter overestimate (since corrected). With litter inputs in the correct boreal range, a multiplicative-normal model is appropriate.

6.2 Fixed Observation Error

σ_{obs} is **fixed** at the observed coefficient of variation of SOC across all calibration plots:

$$\sigma_{\text{obs}} = \frac{\text{sd}(\text{SOC}_{\text{obs}})}{\text{mean}(\text{SOC}_{\text{obs}})} \approx 0.45$$

This value is **identical across all four models**. Fixing σ_{obs} ensures structural differences appear in residuals rather than being absorbed into the error term.

6.3 Log-Likelihood

For each plot i with n_i observations:

$$\ell_i(\theta) = \sum_{k=1}^{n_i} \log \mathcal{N}(\text{SOC}_{i,k};\ \hat{C}_{i,t_{ik}},\ s^2_{i,k})$$

Total log-posterior in unconstrained space:

$$\mathcal{L}(v) = \sum_i \ell_i(\text{to\_original}(v)) + \log \left| \frac{\partial \theta}{\partial v} \right|$$

If any plot returns $-\infty$, the entire proposal is rejected. The inner per-plot function is byte-compiled with `cmpfun()` for a ~10–20% speedup at the R level.

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7 Per-Plot Likelihood Evaluation

Each likelihood evaluation executes three stages per plot in parallel via `mclapply`.

7.1 Stage 1: Environmental Modifier Time Series

For each year t in 1985–2023, `compute_xi()` computes the scalar modifier ξ_t from annual climate summaries (mean temperature, temperature amplitude, annual precipitation). The modifier is applied multiplicatively to all decomposition rates.

7.2 Stage 2: Steady-State Initialisation

The initial SOC pool distribution C^* (5-element vector) is computed analytically:

$$C^* = -A(\theta,\ \xi_{\text{ss}})^{-1}\, b(\theta)$$

where A is the transfer matrix, b is the mean litter input vector, and ξ_{ss} is the climate modifier evaluated at mean climate over the **20-year steady-state window** (1985–2004).

Jensen's inequality correction. Because ξ is nonlinear in T and P:

$$\xi(\bar{T}, \bar{P}) \neq \overline{\xi(T_t, P_t)}$$

Using the average of annual modifiers instead of the modifier at average climate introduces a 5–15% systematic bias in boreal conditions. The correct approach is enforced through the `compute_xi_mean()` interface: it averages T, T_{amp}, and P over the steady-state window and calls `compute_xi()` once.

Litter scaling. σ_{input} is applied to litter inputs in both the steady-state initialisation and the transient run, so the calibrated multiplier affects the initial condition and the trajectory consistently.

7.3 Stage 3: Transient Simulation

Starting from C^*, the model runs forward 1985–2023 using the annual ξ_t time series and actual annual litter inputs. A single Fortran call per plot returns the full pool trajectory; predicted SOC at observation years (1985, 2006) is extracted and compared to observations.

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8 MCMC Sampling

8.1 Sampler

DEzs (Differential Evolution with snooker updates; ter Braak & Vrugt 2008), implemented in `BayesianTools` (Hartig et al. 2023). Selected for:

- No gradient requirements (black-box Fortran model).
- Robust performance on correlated, moderately high-dimensional posteriors (20 free parameters).
- Population-based proposals that efficiently explore posterior ridges from compositional constraints.

8.2 Settings

Setting	Test (Mac)	Production (Puhti)
Chains	4	4
Iterations	10,000	50,000
Burn-in	2,000	10,000
DEzs jitter ε	0.001	0.001
Random seed	2025	2025

Burn-in fraction: 20%. Four chains give meaningfully tighter Gelman–Rubin bounds than three chains for negligible extra cost.

8.3 Parallelisation

Sequential-chains (current): chains run one after another (`lapply`), each using all available cores for the `mclapply` per-plot loop. On Puhti: 39 cores per chain (`parallelly::availableCores()`). On Mac: `detectCores() - 1`.

Hybrid (future Puhti production): outer `lapply` replaced by `mclapply(mc.cores = 4)` with `CORES_PER_CHAIN = 90`, giving $4 \times 90 = 360$ of 384 cores. Nested forking is supported on Linux but not macOS, so sequential is mandatory for Mac development.

`parallel = FALSE` is passed to `createBayesianSetup()` so BayesianTools' built-in parallelism is suppressed.

8.4 Starting Point

All chains start from `best_x = to_unconstrained(published_defaults)`. DEzs initialises its population from small random perturbations around `best_x` and rapidly disperses via differential-evolution proposals.

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9 Pre-MCMC Sanity Check

Before launching MCMC, a forward simulation at the prior centre (published defaults) is run for 4 randomly selected plots:

1. **Numerical stability:** all predicted SOC finite and positive.
2. **Order-of-magnitude check:** predicted / observed $\in [0.1, 10]$ at both observation years.
3. **Likelihood check:** `ll_fn(best_x)` finite across all 350 plots.

Output: `{Model}_forward_at_defaults_{RUN_ID}.png` (2$×$2 panel). If the sanity check fails, MCMC is not launched.

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10 Convergence Diagnostics

All diagnostics computed on post-burnin samples. $\hat{R}$ and ESS computed in unconstrained space; posterior summaries reported in physical space via `to_original()`.

10.1 Gelman–Rubin $\hat{R}$

$\hat{R}$	Interpretation
< 1.05	Good convergence
1.05–1.10	Marginal; extend chains
> 1.10	Poor convergence; run not usable

Computed both per-parameter (univariate, `coda::gelman.diag()`) and jointly (multivariate $\hat{R}$).

10.2 Effective Sample Size

`coda::effectiveSize()`. Values < 200 indicate poor mixing.

10.3 Diagnostic Outputs

All files written to `Calibration_real_data/diagnostics/{Model}/`:

File	Content
`*_traces_*.png`	Trace plots per parameter, one line per chain
`*_marginals_fractions_*.png`	Prior vs posterior KDE for the 12 flow fractions
`*_marginals_climate_nuisance_*.png`	Prior vs posterior KDE for climate and nuisance parameters
`*_rhat_barplot_*.png`	$\hat{R}$ per parameter, coloured by convergence zone
`*_kl_divergence_*.png`	KL(posterior ‖ prior) per parameter
`*_report_*.txt`	$\hat{R}$, ESS, posterior quantile table, chain health

Prior and posterior KDEs are evaluated on the same x-grid derived from the prior's support, not the posterior's range, to avoid misleading compression.

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11 Output Files

Written to `Calibration_real_data/runs/`:

File	Content	Used by
`{Model}_chains_{RUN_ID}.rds`	Raw BayesianTools chain objects (unconstrained)	Chain extension, re-diagnostics
`{Model}_posterior_{RUN_ID}.rds`	Post-burnin samples in **physical space**	Predictive script, downstream analysis
`{Model}_metadata_{RUN_ID}.rds`	Run config, R version, wallclock, summary stats	Reproducibility
`{Model}_inputs_{RUN_ID}.rds`	Per-plot structures: `climate_by_plot`, `inputs_by_plot`, `litter_means`, `obs_meta`, `plot_info`	Predictive script

The inputs RDS ensures exact provenance: the predictive script reads the same per-plot structures the calibration used, even if `Data_work.R` is subsequently updated.

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12 Key Design Decisions

Architectural decisions locked in after discussion (April 2026):

1. **Multiplicative-normal likelihood.** Reverted from a log-normal workaround added when litter inputs had a $\times 10$ overestimate. With correct litter magnitudes, multiplicative-normal is the appropriate model.

2. **Fixed σ_{obs} across all models.** Ensures structural differences appear in residuals. A poorly fitting model cannot mask its inadequacy by calibrating a larger error variance.
3. **$\xi(\bar{T}, \bar{P})$, not $\overline{\xi(T_t, P_t)}$, for steady state.** Jensen's inequality correction; 5–15% bias in boreal conditions. Fixed at the `compute_xi_mean` interface level in `calibration_engine.R`.
4. **STEADY_STATE_YEARS = 20.** Uses 1985–2004 climate for the steady-state scalar, pre-dating accelerated boreal warming. Sensitivity to this choice is a planned post-paper analysis.
5. **Fixed decomposition rates.** $\alpha_A, \alpha_W, \alpha_E, \alpha_N, p_H, \alpha_H$ fixed at published values. Structural intercomparison should be driven by flow fractions and climate response.
6. **Sequential chains.** Mandatory on macOS. Hybrid strategy (4×39 cores) available on Puhti.
7. **Global σ_{input}.** Single litter scaling multiplier. Per-species stratification is a possible extension but would introduce identifiability questions that distract from the intercomparison.

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13 Software

Component	Details
R	$\geq$ 4.3; Puhti: 4.5.2 (`r-env` Apptainer)
`BayesianTools`	Hartig et al. (2023); DEzs sampler
`coda`	Gelman–Rubin $\hat{R}$, ESS
`compiler`	`cmpfun()` byte-compilation
`parallel`	`mclapply()` per-plot parallelism
Fortran model cores	`R CMD SHLIB`; `.Fortran()` interface
HPC	CSC Puhti; `small` partition; 1 node, 40 cores, 2 GB/core

Random seed: `set.seed(2025)` before chain initialisation (consistent across all model scripts). Mac test plot subsampling: `set.seed(42)`.